

DIKSHYA GIRI

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EDUCATION

B.S. Computer Science & Mathematics | Caldwell University

Caldwell, NJ | Expected May 2027

Honors Student • Presidential Scholar | Coursework: Data Structures & Algorithms, AI, Mobile Dev, Probability & Statistics, OOP, DBMS, Full Stack, UI/UX

TECHNICAL SKILLS

Languages & Tools: Python, C/C++, C#, R, SQL, TypeScript, React, Node.js, JavaScript, REST APIs, Docker, Firebase, AWS, Figma

AI/ML: TensorFlow, PyTorch, scikit-learn, Keras, LSTM, RL, RAG, Computer Vision, NLP, YOLO, MediaPipe, SUMO

Data: pandas, NumPy, Matplotlib, Power BI, Tableau, Seaborn, Git/GitHub

AWARDS

1st Place | HackNYU 2025 (650+ participants, 120+ projects)

New York University

2nd Place | HackNJIT 2024 (550+ participants, 75+ projects)

NJ Institute of Technology

Top 10 | CalHacks x Anthropic x Amazon (3,400+ participants, 700+ projects)

UC Berkeley

Finalist | HackMIT x Fetch.ai x Anthropic 2025 (365 projects)

MIT

PROFESSIONAL EXPERIENCE

AI Research Intern | Smart Traffic Signal Control | Caldwell University

Caldwell, NJ | May – Sep 2025

Built an end-to-end adaptive traffic system using computer vision and reinforcement learning to replace fixed-timing signals

- Engineered a real-time vehicle detection pipeline with OpenCV achieving 85% accuracy at 15 FPS, replacing a legacy fixed-timing system used across 4 traffic load profiles.
- Trained a reinforcement learning signal controller in SUMO simulations that cut average vehicle wait time by 15.7% , outperforming every fixed-timing baselines.
- Reduced vehicle count prediction error from 31% to 18% MAE by designing an LSTM forecasting model, enabling proactive signal adjustment before congestion builds.

Project Lead | Dristi (1st Place, HackNYU 2025) | Google × NYU

Brooklyn, NY | Feb 2025

Led a team building an AI wearable for visually impaired users with real-time obstacle detection and voice navigation

- Architected the full detection system using OpenCV and MediaPipe, re-engineered the pipeline with multi-threading and shared-memory buffering, boosting throughput by 80%.
- Built a Firebase + Cloud Functions backend for emergency alerts, cutting end-to-end latency from ~2s to ~1.1s (45% reduction).

Software Engineering Intern | Lura Health

Newark, NJ | June – Aug 2024

NSF-backed health-tech startup developing a next-generation wearable health monitoring device

- Developed a React + Node.js internal dashboard with JWT authentication and REST API integration, enabling secure physician access to AI diagnostic model outputs.
- Improved health-insight retrieval accuracy by 15% over keyword search across 2000+ patient records by building a RAG recommendation engine with OpenAI embeddings, containerized with Docker for scalable deployment.
- Increased sensor data pipeline throughput by 30% and reduced packet loss by tuning frame sizes and buffer allocation, directly improving reliability of continuous patient health readings.

SELECTED PROJECTS

CONCAVE | Autonomous Smart Agriculture Robot

San Francisco, CA | Nov 2025

- Built an autonomous robot pipeline with ROS, GPS mapping, and OpenCV-based field scanning; adaptive path planning increased field-coverage efficiency by 40% over fixed-path baselines.
- Trained a PyTorch model for ripe-orange detection and early crop disease identification, reaching 95% accuracy and cutting manual inspection time by 60% in field trials.

Gamify.work() | Gesture-Controlled Robotic Arm System.

Newark, NJ | Nov 2024

- Built a real-time hand tracking system using OpenCV, MediaPipe, and TensorFlow that maps live gestures to robotic arm commands over Bluetooth with sub-100ms end-to-end latency.
- Developed a PyGame interface for remote industrial task execution, enabling precise robotic actuation from gesture input alone with no hardware controller required.